Two ray ground reflected propagation

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This derivation of the two ray propagation problem will follow an approach that uses the assumption that the separation distance between TX and RX is very large compared to the height of both TX and RX ($d >> h_{TX}$ and $d >> h_{RX}$). This produces a simplified result, where the ground reflection coefficient is taken to be -1 and the two waves are added coherently (not in phase, but parallel in direction).

Next the derivation is extended to include a ground reflection coefficient.

Finally, an approach is derived using Poynting vectors where the propagation direction of each wave is taken into account.

1 Geometry

The geometric situation is for $h_{TX} > h_{RX}$ described in Figure 1.

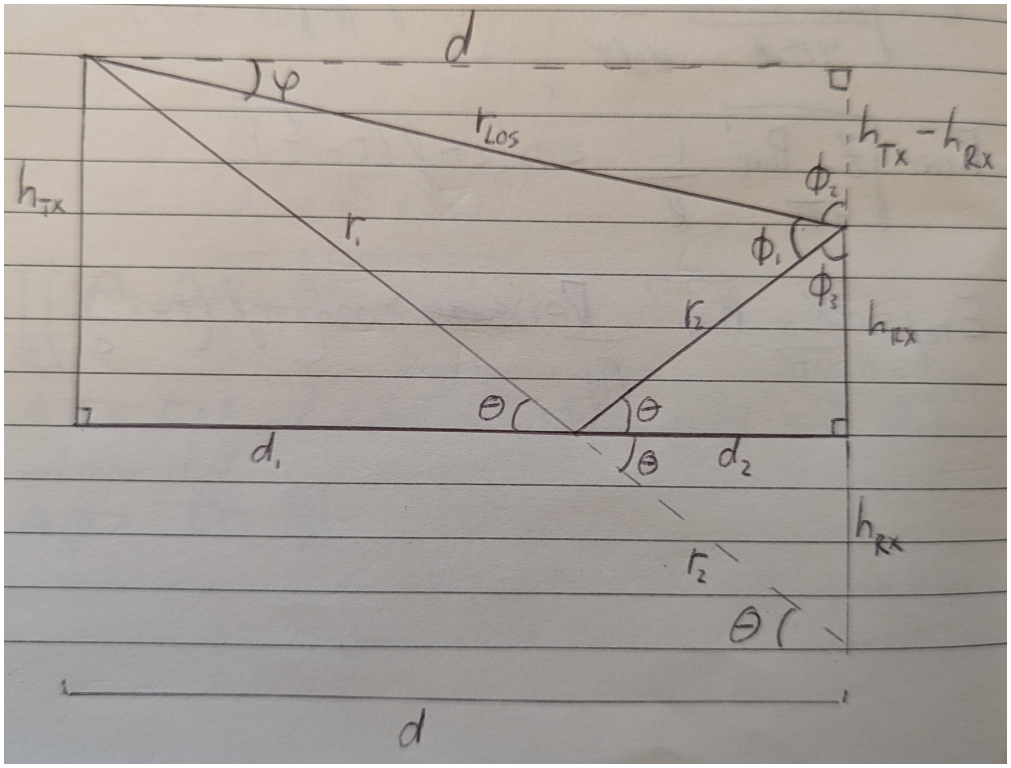

Figure 1: Geometry of the two ray ground reflected propagation problem ($h_{TX} > h_{RX}$)

The variables h_{TX}, h_{RX} and d are known beforehand. The following two variables can then be derived :

$$r_{LOS} = \sqrt{d^2 + (h_{TX} - h_{RX})^2} \tag{1}$$

$$r_{ref} = \sqrt{d^2 + (h_{TX} + h_{RX})^2} \tag{2}$$

In the more thorough derivation, the following variables are needed:

$$\theta = \arctan\left(\frac{h_{TX} + h_{RX}}{d}\right) \tag{3}$$

$$\begin{aligned}
\phi_1 &= 180^o - \phi_2 - \phi_3 = 180^o - \arctan\left(\frac{d}{h_{TX} - h_{RX}}\right) - (180^o - 90^o - \theta) \\
&= 90^o - \arctan\left(\frac{d}{h_{TX} - h_{RX}}\right) + \theta \\
&= 90^o - \arctan\left(\frac{d}{h_{TX} - h_{RX}}\right) + \arctan\left(\frac{h_{TX} + h_{RX}}{d}\right) \\
&= \frac{\pi}{2} - \arctan\left(\frac{d}{h_{TX} - h_{RX}}\right) + \arctan\left(\frac{h_{TX} + h_{RX}}{d}\right)
\end{aligned} \tag{4}$$

Notice that if $h_{TX} = h_{RX}$, the equation above no longer holds due to a division by zero error. From the image we can see that $\phi_2 = \frac{\pi}{2}$, from which follows that $\phi_1 = \theta$.

If $h_{TX} < h_{RX}$ we get another scenario described in Figure 2. The values for θ, r_{LOS} and r_{ref} remain

the same as in the $h_{TX} > h_{RX}$ case (and in the $h_{TX} = h_{RX}$ for that matter). The derivation of ϕ_1 according to Figure 2 yields:

$$
\begin{aligned}
\phi_1 &= 90^o - \phi_2 - \phi_3 \\
&= 90^o - \arctan\left(\frac{h_{RX} - h_{TX}}{d}\right) - (90^o - \theta) \\
&= \theta - \arctan\left(\frac{h_{RX} - h_{TX}}{d}\right) \\
&= \arctan\left(\frac{h_{TX} + h_{RX}}{d}\right) - \arctan\left(\frac{h_{RX} - h_{TX}}{d}\right)
\end{aligned} \tag{5}
$$

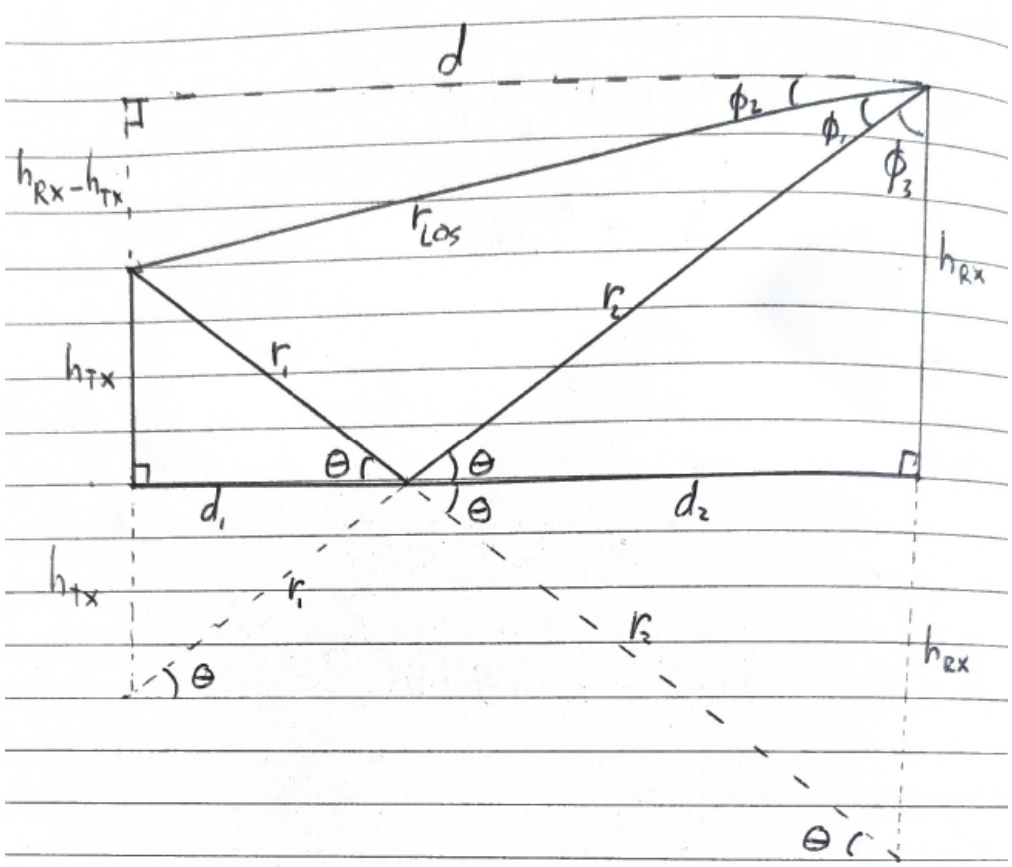

Figure 2: Geometry of the two ray ground reflected propagation problem ($h_{TX} < h_{RX}$).

2 Basic derivation

We start with the E-field equations of both propagation paths. For the LOS path we have:

$$E_{LOS}(t) = \frac{E_0}{r_{LOS}} \exp\left(i\omega(\frac{r_{LOS}}{c} - t)\right) \tag{6}$$

For the reflected path we have:

$$E_{ref}(t) = -\frac{E_0}{r_{ref}} \exp\left(i\omega(\frac{r_{ref}}{c} - t)\right) \tag{7}$$

The minus sign comes from the reflection coefficient, which is assumed to be -1 for small scraping angles. In the case that $d >> h_{TX}$ and $d >> h_{RX}$, the rays are almost parallel and we assume that we can coherently add the two waves together:

$$E_{tot}(t) \approx \frac{E_0}{r_{LOS}} \exp\left(i\omega(\frac{r_{LOS}}{c} - t)\right) - \frac{E_0}{r_{ref}} \exp\left(i\omega(\frac{r_{ref}}{c} - t)\right) \tag{8}$$

Next we evaluate the situation at time $t = r_{LOS}/c$. We take the substitution $r_\Delta = r_{ref} - r_{LOS}$.

$$E_{tot}(t = \frac{r_{LOS}}{c}) \approx \frac{E_0}{r_{LOS}} - \frac{E_0}{r_{ref}} \exp\left(i\frac{\omega r_\Delta}{c}\right) \tag{9}$$

Again following the assumption that $d >> h_{TX}$ and $d >> h_{RX}$, the relative path lengths are almost equal $r_{LOS} \approx r_{ref} \approx d$ and only the phase difference is considered:

$$\begin{aligned}
E_{tot}(t = \frac{r_{LOS}}{c}) &\approx \frac{E_0}{d}\left(1 - \exp\left(i\frac{\omega r_\Delta}{c}\right)\right) && (10)\\
&= \frac{E_0}{d}\left(1 - \cos\left(\frac{\omega r_\Delta}{c}\right) - i\sin\left(\frac{\omega r_\Delta}{c}\right)\right) && (11)\\
&= \frac{E_0}{d}\left(1 - \cos\left(\frac{\omega r_\Delta}{c}\right)\right) - i\frac{E_0}{d}\sin\left(\frac{\omega r_\Delta}{c}\right) && (12)
\end{aligned}$$

We get the magnitude of E_{tot} using $|E_{tot}| = \sqrt{Re(E_{tot})^2 + Im(E_{tot})^2}$:

$$\begin{aligned}
|E_{tot}| &\approx \frac{E_0}{d}\sqrt{\left(1 - \cos\left(\frac{\omega r_\Delta}{c}\right)\right)^2 + \sin^2\left(\frac{\omega r_\Delta}{c}\right)} && (13)\\
&= \frac{E_0}{d}\sqrt{2 - 2\cos\left(\frac{\omega r_\Delta}{c}\right)} && (14)\\
&= \frac{E_0}{d}\sqrt{2\left(1 - \cos\left(\frac{\omega r_\Delta}{c}\right)\right)} && (15)\\
&= \frac{E_0}{d}\sqrt{4\sin^2\left(\frac{\omega r_\Delta}{2c}\right)} && (16)\\
&= \frac{E_0}{d}2\sin\left(\frac{\omega r_\Delta}{2c}\right) && (17)
\end{aligned}$$

By writing out r_Δ we get:

$$\begin{aligned}
r_\Delta &= r_{ref} - r_{LOS} && (18)\\
&= \sqrt{d^2 + (h_{TX} + h_{RX})^2} - \sqrt{d^2 + (h_{TX} - h_{RX})^2} && (19)\\
&= d\left(\sqrt{1 + \frac{(h_{TX} + h_{RX})^2}{d^2}} - \sqrt{1 + \frac{(h_{TX} - h_{RX})^2}{d^2}}\right) && (20)
\end{aligned}$$

And substituting equation 19 into equation 17 we get:

$$|E_{tot}| = \frac{E_0}{d} 2 \sin\left(\frac{\omega}{2c}\left(\sqrt{d^2 + (h_{TX} + h_{RX})^2} - \sqrt{d^2 + (h_{TX} - h_{RX})^2}\right)\right) \tag{21}$$

The first step to simplifying this is by rewriting r_Δ with the assumption that $d >> h_{TX}$ and $d >> h_{RX}$. We can then use the Taylor approximation $\sqrt{1+x} \approx 1 + x/2$ for small x in equation 20:

$$\begin{aligned}
r_\Delta &\approx d\left(1 + \frac{(h_{TX} + h_{RX})^2}{2d^2} - 1 - \frac{(h_{TX} - h_{RX})^2}{2d^2}\right) && (22)\\
&= \frac{1}{2d}\left(h_{TX}^2 + h_{RX}^2 + 2h_{TX}h_{RX} - h_{TX}^2 - h_{RX}^2 + 2h_{TX}h_{RX}\right) && (23)\\
&= \frac{4h_{TX}h_{RX}}{2d} && (24)\\
&= \frac{2h_{TX}h_{RX}}{d} && (25)
\end{aligned}$$

To further simplify equation 21 we can use the small angle approximation $\sin(\theta) \approx \theta$ if θ is very small. To do that, we have to make sure that $\frac{\omega r_\Delta}{2c}$ is very small. We use $\omega = 2\pi f = 2\pi c/\lambda$ to get $\frac{\omega r_\Delta}{2c} = \frac{\pi r_\Delta}{\lambda}$. Again under the assumption $d >> h_{TX}$ and $d >> h_{RX}$, r_Δ is small and we use the small angle approximation (as long as the wavelength is also long enough):

$$\begin{aligned}
|E_{tot}| &\approx \frac{E_0}{d} 2 \sin\left(\frac{\pi r_\Delta}{\lambda}\right) && (26)\\
&\approx \frac{E_0}{d}\frac{2\pi}{\lambda} r_\Delta && (27)\\
&\approx \frac{E_0}{d}\frac{2\pi}{\lambda}\frac{2h_{TX}h_{RX}}{d} && (28)\\
&\approx \frac{E_0 4\pi h_{TX}h_{RX}}{d^2\lambda} && (29)
\end{aligned}$$

The power density S_{RX} at RX is then:

$$S_{RX} = \frac{|E_{tot}|^2}{Z_0} \approx \frac{1}{Z_0}\left(\frac{E_0 4\pi h_{TX}h_{RX}}{d^2\lambda}\right)^2 \tag{30}$$

The amplitude E_0 can be derived using the Poynting vector $S = E \times H$ and using the plane wave approximation $H \approx E/Z_0$, where Z_0 is the free space impedance, to give:

$$E_0 = \sqrt{\frac{P_{EIRP} Z_0}{4\pi}} \tag{31}$$

Combining it all:

$$S_{RX} \approx 4\pi\left(\frac{h_{TX}h_{RX}}{\lambda}\right)^2 \frac{1}{d^4} P_{EIRP} = p(d) P_{EIRP} \tag{32}$$

As can be seen in the last section, the two ray model is plotted in Figure 5, there is a $1/r^2$ and a $1/r^4$ part. The free space propagation path is the average of the $1/r^2$ part. Equation the free space propagation function with equation 32, we get the transition distance r_b:

$$p_{FS}(r_b) = p_{2-ray}(r_b) \tag{33}$$

$$\frac{1}{4\pi r_b^2} = 4\pi\Big(\frac{h_{TX}h_{RX}}{\lambda r_b^2}\Big)^2 \tag{34}$$

$$r_b = \frac{4\pi h_{TX}h_{RX}}{\lambda} \tag{35}$$

This is consistent with the derivation by Seybold [1, p. 172], but Molisch seems to have forgotten a factor π [2].

3 Extended derivation: including ground reflection coefficient

The two waves that include the reflection coefficient are as follows. For the LOS path we have again Equation 6 and for the reflected path we have:

$$E_{ref}(t) = \Gamma(\theta)\frac{E_0}{r_{ref}}\exp\Big(i\omega(\frac{r_{ref}}{c} - t)\Big) \tag{36}$$

Where the reflection coefficient $\Gamma(\theta)$ for a perpendicular polarization is given by [3] (see Figure 1 or 2 for θ):

$$\Gamma_{\perp}(\theta) = \frac{\sin(\theta) - \sqrt{\epsilon_r - \cos^2(\theta)}}{\sin(\theta) + \sqrt{\epsilon_r - \cos^2(\theta)}} \tag{37}$$

$$\Gamma_{||}(\theta) = \frac{\sqrt{\epsilon_r}\sin(\theta) - \sqrt{\epsilon_r - \cos^2(\theta)}}{\sqrt{\epsilon_r}\sin(\theta) + \sqrt{\epsilon_r - \cos^2(\theta)}} \tag{38}$$

Complex reflection coefficients are derived by Jordan in [4, p. 631-632] and are given below:

$$\Gamma_{\perp}(\theta) = \frac{\sin(\theta) - \sqrt{y - \cos^2(\theta)}}{\sin(\theta) + \sqrt{y - \cos^2(\theta)}} \tag{39}$$

$$\Gamma_{||}(\theta) = \frac{y\sin(\theta) - \sqrt{y - \cos^2(\theta)}}{y\sin(\theta) + \sqrt{y - \cos^2(\theta)}} \tag{40}$$

where

$$y = \epsilon_r - ix \tag{41}$$

$$x = 18 * 10^9 * \frac{\sigma}{f} \tag{42}$$

$$\tag{43}$$

and σ is the ground conductivity.

Adding both LOS an reflected waves under the assumption that both waves can be coherently added together we get:

$$E_{tot}(t) \approx \frac{E_0}{r_{LOS}} \exp\left(i\omega(\frac{r_{LOS}}{c} - t)\right) + \Gamma(\theta)\frac{E_0}{r_{ref}} \exp\left(i\omega(\frac{r_{ref}}{c} - t)\right) \tag{44}$$

As in the derivation before we look at $t = \frac{r_{LOS}}{c}$ and assume $r_{LOS} \approx r_{ref} \approx d$ and get:

$$E_{tot}(t = \frac{r_{LOS}}{c}) \approx \frac{E_0}{d}\left(1 + \Gamma(\theta)\exp\left(i\omega(\frac{r_\Delta}{c})\right)\right) \tag{45}$$

$$= \frac{E_0}{d}\left(1 - \Gamma(\theta)\cos\left(\frac{\omega r_\Delta}{c}\right)\right) - i\frac{E_0}{d}\Gamma(\theta)\sin\left(\frac{\omega r_\Delta}{c}\right) \tag{46}$$

The magnitude of E_{tot} is then $|E_{tot}| = \sqrt{Re(E_{tot})^2 + Im(E_{tot})^2}$:

$$|E_{tot}| = \frac{E_0}{d}\sqrt{\left(1 - \Gamma^2(\theta)\cos\left(\frac{\omega r_\Delta}{c}\right)\right)^2 + \Gamma^2(\theta)\sin^2\left(\frac{\omega r_\Delta}{c}\right)} \tag{47}$$

And the power density at RX is then:

$$S_{RX} = \frac{|E_{tot}|^2}{Z_0} = \frac{E_0^2}{Z_0 d^2}\left(\left(1 - \Gamma^2(\theta)\cos\left(\frac{\omega r_\Delta}{c}\right)\right)^2 + \Gamma^2(\theta)\sin^2\left(\frac{\omega r_\Delta}{c}\right)\right) \tag{48}$$

4 (More) thorough derivation results

CHANGE TO ALIGN WITH COORDINATE SYSTEM OF KRAUS AND THE REFLECTION COEFFICIENTS

To calculate the two-ray model numerically, we are not bound by easy to read simplifications and approximations to get an intuition of the situation. Here the computer does all the heavy lifting for us and we can implement a more thorough approach.

First we start by defining the reflection coefficient for perpendicular $r\Gamma_\perp$ and parallel $\Gamma_{||}$ polarised waves, according to [3, p. 456-458], as:

$$\Gamma_\perp = \frac{\cos(\theta_\Gamma) - \sqrt{\epsilon_r - \sin^2(\theta_\Gamma)}}{\cos(\theta_\Gamma) + \sqrt{\epsilon_r - \sin^2(\theta_\Gamma)}} \tag{49}$$

$$\Gamma_{||} = \frac{-\sqrt{\epsilon_r}\cos(\theta_\Gamma) + \sqrt{\epsilon_r - \sin^2(\theta_\Gamma)}}{\sqrt{\epsilon_r}\cos(\theta_\Gamma) + \sqrt{\epsilon_r - \sin^2(\theta_\Gamma)}} \tag{50}$$

, where $\theta_\Gamma = \frac{\pi}{2} - \theta = 90^o - \theta$ and θ is as defined in Figure 1 and 2.

At the receiver, we want to know the electric and magnetic field components of both direct and reflected waves. We consider the reflection coefficient and the difference in angle of incidence at the receiver, but assume both waves arrive as plane waves. In this case we can no longer simply add the two waves coherently, but have to keep the vector components of the fields in mind for both waves when

calculating the power of the combined waves at the receiver using the Poynting vector equation. The Poynting vector equation is given by:

$$\begin{aligned} \boldsymbol{S} &= \boldsymbol{E} \times \boldsymbol{H} \\ &= (\boldsymbol{E}_{LOS} + \boldsymbol{E}_{ref}) \times (\boldsymbol{H}_{LOS} + \boldsymbol{H}_{ref}) \end{aligned} \tag{51}$$

This form of the Poynting vector equation hold when there are no free charges or currents in the medium in which the waves travel and the medium behaves linearly, which is true considering waves traveling in free space.

In Figure 3 the electric field vectors and the wave vectors of the LOS and reflected waves can be seen. The magnetic components are perpendicular to both these vectors.

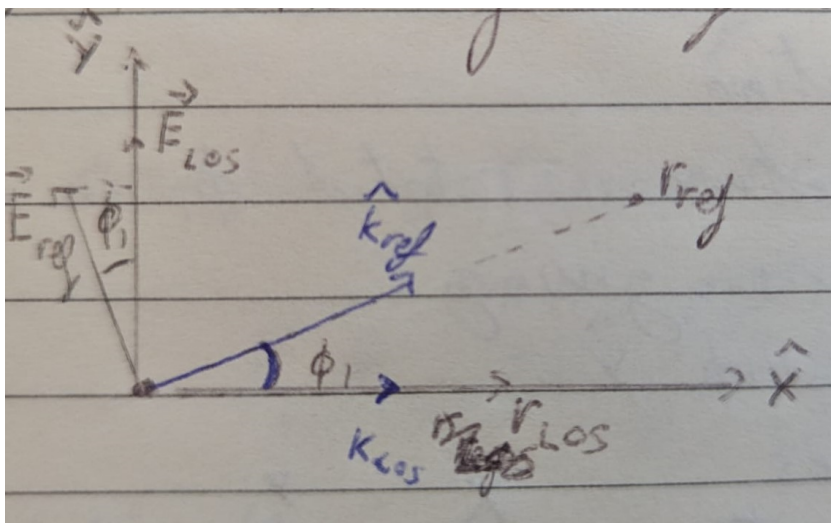

Figure 3: Electric field vectors and wave vectors of the LOS and reflected waves at RX.

We will see later that the magnetic field vectors can be easily calculated from the electric field vectors in the plane wave assumption. Therefore, we first need to derive $\boldsymbol{E_{LOS}}$ and $\boldsymbol{E_{ref}}$.

The general E-field description of a wave is given by:

$$\boldsymbol{E}(\boldsymbol{r}, t) = \frac{E_0}{r} \exp\Big(- i(\omega t - \boldsymbol{k} \cdot \boldsymbol{r})\Big)\boldsymbol{n} \tag{52}$$

, where E_o is the amplitude constant (given in equation 31), $\omega = 2\pi f$ is the angular frequency, t is the time, $\boldsymbol{k}$ is the wave vector ($|\boldsymbol{k}| = \frac{2\pi}{\lambda}$) and $\boldsymbol{n}$ is the direction of the E-field.

4.1 $\boldsymbol{E_{LOS}}$

As we are working with vectors we need to pick an appropriate coordinate system. A suitable choice is that of Figure 3, which in the context of Figure 1 and 2 defines the axes as seen in Figure 4.

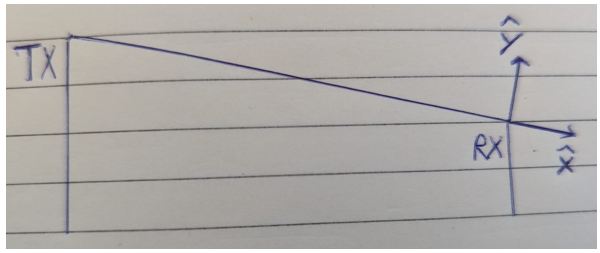

Figure 4: Coordinate system used for the calculations. The z axis comes out of the page.

In this coordinate system we start by constructing perpendicular and parallel polarised waves from equation 52 as:

$$\boldsymbol{E}_{\perp,LOS} = \frac{E_0}{r_{LOS}} \exp\Big(- i2\pi(ft - \frac{r_{LOS}}{\lambda})\Big)\hat{\boldsymbol{z}} \tag{53}$$

$$\boldsymbol{E}_{||,LOS} = \frac{E_0}{r_{LOS}} \exp\Big(- i2\pi(ft - \frac{r_{LOS}}{\lambda})\Big)\hat{\boldsymbol{y}} \tag{54}$$

Note that the vectors $\boldsymbol{k}$ and $\boldsymbol{r}$ are aligned, such that the dot product in equation 52 simply becomes a multiplication of their absolute values.

For plane waves in free space the corresponding magnetic field vectors are given by $|\boldsymbol{H}| = \frac{|\boldsymbol{E}|}{Z_0}$ and thus:

$$\boldsymbol{H}_{\perp,LOS} = \frac{E_0}{Z_0 r_{LOS}} \exp\Big(- i2\pi(ft - \frac{r_{LOS}}{\lambda})\Big)\hat{-\boldsymbol{y}} \tag{55}$$

$$\boldsymbol{H}_{||,LOS} = \frac{E_0}{Z_0 r_{LOS}} \exp\Big(- i2\pi(ft - \frac{r_{LOS}}{\lambda})\Big)\hat{\boldsymbol{z}} \tag{56}$$

4.2 $\boldsymbol{E_{ref}}$

The derivation of $\boldsymbol{E_{ref}}$ is a little less straightforward. It differs from $\boldsymbol{E_{LOS}}$ in (a) the distance $r = r_{ref}$, (b) the direction of propagation, (c) the direction of the E and H fields and (d) the reflection attenuation Γ.

(a) is straightforward by considering r_{ref} instead of r_{LOS}.

(b) The direction of propagation is contained in the wave vector $\boldsymbol{k}$. We can say that $\boldsymbol{r}$ and $\boldsymbol{k}$ are aligned by imagining that the reflected wave traveled in a straight line with length r_{ref} and the reflection is simply a multiplication of this wave by an attenuation factor. So both $\boldsymbol{r}$ and $\boldsymbol{k}$ have changed direction in the same way and the exponent in equations 53 and 54 remains the same.

(c) The direction of the fields changes at the hand of Figure 3. The field vectors for the reflected wave is adjusted by breaking it up in x and y components using the angle ϕ_1. That means that $\boldsymbol{k}_{ref}$ is given by:

$$\boldsymbol{k}_{ref} = \cos(\phi_1)\boldsymbol{x} + \sin(\phi_1)\boldsymbol{y} \tag{57}$$

The directions of the E and H fields must thus be adjusted correspondingly.

(d) Here we simply multiply the equations by a reflection attenuation factor Γ from equations 49 and 50.

Combining points (a) through (d) we have the following expressions for the reflected wave:

$$\boldsymbol{E}_{\perp,ref} = \frac{E_0}{r_{ref}}\Gamma_\perp \exp\left(-i2\pi(ft - \frac{r_{ref}}{\lambda})\right)\hat{\boldsymbol{z}} \tag{58}$$

$$\boldsymbol{E}_{||,ref} = \frac{E_0}{r_{ref}}\Gamma_{||} \exp\left(-i2\pi(ft - \frac{r_{ref}}{\lambda})\right)(\cos(\phi_1)\hat{\boldsymbol{y}} - \sin(\phi_1)\hat{\boldsymbol{x}}) \tag{59}$$

$$\boldsymbol{H}_{\perp,ref} = \frac{E_0}{Z_0 r_{ref}}\Gamma_\perp \exp\left(-i2\pi(ft - \frac{r_{ref}}{\lambda})\right)(\sin(\phi_1)\hat{\boldsymbol{x}} - \cos(\phi_1)\hat{\boldsymbol{y}}) \tag{60}$$

$$\boldsymbol{H}_{||,ref} = \frac{E_0}{Z_0 r_{ref}}\Gamma_{||} \exp\left(-i2\pi(ft - \frac{r_{ref}}{\lambda})\right)\hat{\boldsymbol{z}} \tag{61}$$

4.3 Power at RX

The equations for $\boldsymbol{E}_{\perp,LOS}, \boldsymbol{E}_{||,LOS}, \boldsymbol{H}_{\perp,LOS}, \boldsymbol{H}_{||,LOS}, \boldsymbol{E}_{\perp,ref}, \boldsymbol{E}_{||,ref}, \boldsymbol{H}_{\perp,ref}, \boldsymbol{H}_{||,ref}$ can then be plugged into equation 51 to get the poynting vector for parallel and perpendicular components. The derivation stops here as showing this final step is quite cumbersome and yields no real gain. The derivation so far will let you generate a computer script that calculates the two-ray model in a less simplified manner.

This script has been made and Figure 5 shows the power density for parallel polarization at different relative permittivities.

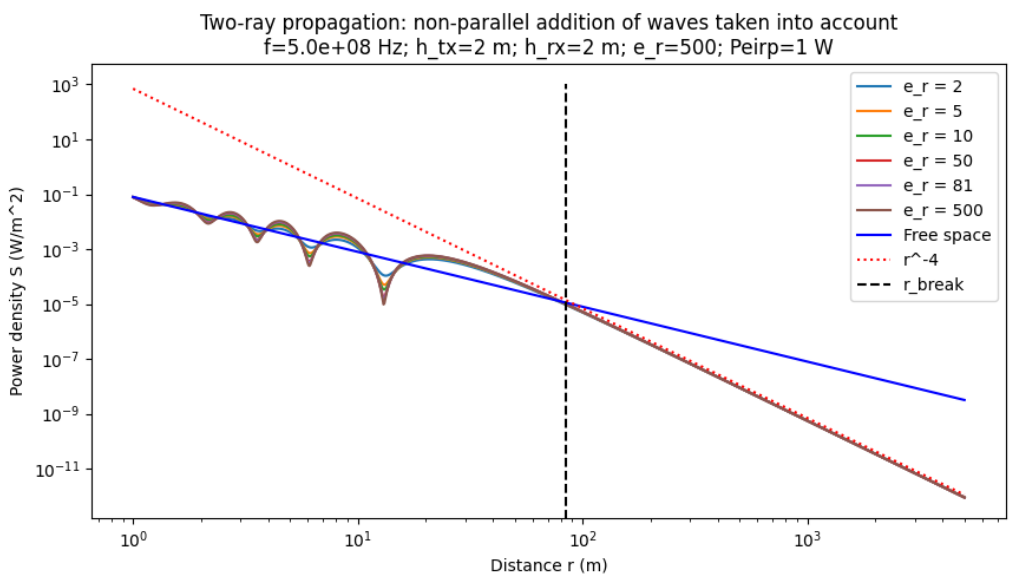

Figure 5: Power density at RX with respect to the separation distance d of a two-ray ground reflected propagation model.

In Figure 5 the $1/d^4$ region is independent on the relative permeability ϵ_r. The book by Molisch says that measurements at longer distances have a $1/d^6$ dependency [2]. This is not at all captured by this model on a flat earth.

References

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